NOT MEASUREMENT SENSITIVE

MIL-P-85856/11
7 December 1988

MILITARY SPECIFICATION SHEET

POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS
PURE NICKEL, TYPE I, CLASS 1, COMPOSITION J

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

Chemical Composition

Ni	Fe	Co	Others (1)	
			Each	Total
Bal	0.3 max.	0.3 max.	-	0.2 max.

(1) Determination not required for routine acceptance.

Particle Size/Distribution

+325 Mesh (+45 µm)	5.0 max.
-325 Mesh (-45 µm)	95.0 min.
-10 µm	5.0 max.

Sprayed Coating Properties

Bond Strength	21.0 MPa (3000 psi) min.
Hardness	Rockwell 15T, 80 min.
Microstructure	See paragraph 4.5.4.1.2 Porosity - 15% max. Unmelted Particles - 15% max.

Deposit Efficiency and Deposition Rate

65% min for N_2/H_2 @ 15.0 lb/hr
55% min for A/H_2 @ 15.0 lb/hr

AMSC N/A FSC 3439

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Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

General Electric Company	B50TF17, Class A
Pratt & Whitney Aircraft	PWA 1324

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/11)

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